Physical-Energy Certificates for BKT Gaussian Interfaces: Finite Pins, Charged Sources, and Cell Penalties

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M397

October 9, 2026

Abstract

We study the Gaussian interfaces in the cached manuscript *BKT universality for height and planar spin fields*, without certifying its critical roughening, renormalisation, or spin-field claims. For two separated pin spaces, an orthogonal finite-probe compression admits a two-sided determinant enclosure proportional to the squared missing Hilbert–Schmidt norm. A nonzero source does not obey that quadratic error rule: we derive its exact first-order correction and a signed quadratic residual enclosure, and give a counterexample to discarding the correction. For a separately specified, area-weighted full-cell mass penalty, we obtain an explicit covariance comparison and prove a joint continuum cell-refinement/hard-band limit without a relative-rate condition. Physical Gram matrices, released constants, and paired source elimination are retained throughout. Classical Gaussian conditioning, singular-value minimax, and variational convergence are credited as tools; the increment is their quantitative application to these particular interfaces. Contribution assessment: C3/5, explicit pin-tail and charged-source certificates and a joint continuum penalty limit provide notable local improvements, but do not establish BKT universality.

Contents

1 Problem, conventions, and proof boundary **2**

2 The physical Gaussian pin geometry **2**

3 A quadratic finite-probe determinant certificate **3**

4 Charged sources: the first-order term cannot be discarded **4**

5 Eliminate the source and the boundary together **5**

6 An area-weighted cell-penalty extension **6**

7 Source architecture and retained obligations **8**

8 Comparison, current literature, and practical limits **9**

9 Verification and contribution assessment **10**

1 Problem, conventions, and proof boundary

The primary object is the September 24, 2026 manuscript [1]. Its height theorem concerns admissible finite square-symmetric interactions containing nearest neighbours, and its spin theorems concern the ordinary nearest-neighbour Villain and XY models at a fixed sufficiently large inverse temperature. Those spin theorems are not critical-spin statements. The source's roughening endpoint and all its renormalisation estimates remain source claims in this report. Reading their proofs is not an independent certification of them.

For the height model the source uses heights in $2\pi\mathbb{Z}$, $v_J^2 = (2|J|)^{-1}\sum_{y\in J} y_1^2$, normalized bare parameter $k = \beta/v_J^2$, and continuum covariance $a(k)(-\Delta)^{-1}$. Thus $\beta_{\text{eff}} = v_J^2 a$; for nearest neighbours $v_J^2 = 1/4$. Its claimed critical value is $a = 8\pi$, not a change of convention to $\beta_{\text{eff}} = 8\pi$. In the low-temperature Villain bridge, $\beta = \pi^2 b$ and $K_{\text{V}} = \beta_{\text{eff}}/\pi^2$. For XY, the compensated parameters instead give $K_{\text{XY}} = b/[s + (1+s_{12})^2/(1+s_{11})]$. The mixed compensated XY Gaussian is not a positive joint probability law. None of our positive-Hilbert-space arguments is applied to that complex prescription.

The spin graph Laplacian is unscaled, with one term per unoriented edge and fixed outer angles zero. A signed insertion l need not be neutral: the fused boundary supplies its compensating divergence. With G_n the inverse killed graph Laplacian, the source's full-variance normalization is

$$A_M(b)e^{G_n(x,x)/(2K_M(b))}e^{i\theta_x}.$$

It keeps the finite spatial part of the diagonal. By contrast, the continuum Gaussian pin geometry below has free outer heights and a mass penalty confining its constants. Fixed spin angles do not impose an unmentioned Dirichlet condition on this different height problem.

2 The physical Gaussian pin geometry

Let U be a bounded Lipschitz planar domain, or finitely many such components, and let $a > 0$. Consider on a closed subspace of $H^1(U)$ the real symmetric form

$$T(u,v) = a^{-1}\int_U \nabla u\cdot\nabla v + \langle Bu, QBv\rangle, \qquad Q \geq 0. \tag{1}$$

Here B is a fixed finite list of bounded density observations. Impose only the hard pins actually present, and assume T is coercive on the chosen space: every released constant is either confined by B, Q or explicitly quotiented with compatible neutral loads. The completion in T is the physical energy space $\mathcal{E}$; write its inner product as $\langle\cdot,\cdot\rangle_{\mathcal{E}}$. All projections below are orthogonal in this metric, not in arbitrary coefficient coordinates.

In the source's fixed-cell Gaussian-localization lemma the mass is $\sum_d(\text{avg}_d\, u)^2$, with coefficient one and cell areas bounded below. This is not the L^2 projection mass $\sum_d |d|(\text{avg}_d\, u)^2$ used in our refinement extension below. Both fit (1), with different Q. Fixed-cell constants, fixed observation rank, and the source's Hilbert–Schmidt estimates must not be assumed uniform when the cells or these weights change.

For two disjoint positive-area pin regions H, p, let

$$V_H = \{u\in\mathcal{E} : u = 0 \text{ a.e. on } H\}^{\perp}, \qquad V_p = \{u\in\mathcal{E} : u = 0 \text{ a.e. on } p\}^{\perp}.$$

Set $C = \Pi_H|_{V_p} : V_p \to V_H$. Our standing analytic premises are

$$C\in\mathfrak{S}_2, \qquad \|C\| \leq \kappa < 1. \tag{2}$$

The source's Proposition `an:gaussian-localization` proves these in its fixed permitted geometries using harmonic localization, a cutoff angle bound, and a summable singular-value tail. We do not replace those geometric estimates by finite matrix tests. The spectral gap alone does not imply a determinant, and a bound on operator-norm convergence does not imply convergence of the determinants.

The centered full-pin interaction, interpreted as a limit of finite density ratios, is

$$D(C) = -\tfrac{1}{2}\log\det(I - C^*C) = \tfrac{1}{2}\sum_{j\geq 1} -\log(1-\sigma_j(C)^2). \tag{3}$$

It is finite by (2). This is the logarithm of the joint zero-pin density divided by the product of the two marginal zero-pin densities, not a density of a continuum field at a point.

3 A quadratic finite-probe determinant certificate

Theorem 3.1 (Orthogonal compression enclosure)**.** *Let P and Q_0 be arbitrary finite-rank orthogonal projections on V_H and V_p, respectively, and set $C_0 = PCQ_0$, extended by zero. Then, for*

$$e = \|C - C_0\|_{\mathrm{HS}}^2 = \|C\|_{\mathrm{HS}}^2 - \|C_0\|_{\mathrm{HS}}^2, \tag{4}$$

one has

$$\frac{e}{2} \leq D(C) - D(C_0) \leq \frac{e}{2(1-\kappa^2)}. \tag{5}$$

The ranks need not agree. If only a certified bound $e \leq \bar{e}$ is known, the corresponding computable enclosure is $D(C_0) \leq D(C) \leq D(C_0) + \bar{e}/[2(1-\kappa^2)]$; a positive lower correction requires a lower bound on e as well.

Proof. The map $X \mapsto PXQ_0$ is an orthogonal projection in the Hilbert space of Hilbert–Schmidt operators, proving (4) by Pythagoras. The approximation-number characterization of singular values gives $\sigma_j(PCQ_0) \leq \sigma_j(C)$: multiply a rank-less-than-j approximation to C by the two contractions, and take the infimum. Pad the finite singular-value list by zeros. For $0 \leq y \leq x \leq \kappa^2$,

$$x - y \leq -\log(1-x) + \log(1-y) \leq (x-y)/(1-\kappa^2).$$

Apply this to the squared singular values, sum, and use their trace sum difference e. Absolute convergence follows from the same upper bound, which also justifies the infinite-dimensional passage. □

This is a refinement of the source's qualitative trace approximation, not a new general singular-value theorem. Its constants are sharp given only e and the cap: take a single omitted singular value tending to zero for the lower coefficient, and compare a retained singular value near κ with a slightly smaller one using a slightly tilted rank-one projection for the upper coefficient. The latter compression is realizable by projecting the left singular vector onto a nearby line.

Proposition 3.2 (Finite numerical error is a separate term)**.** *If $\widetilde{C}_0$ is a computed finite matrix in orthonormal physical coordinates, both norms are at most $\bar{\kappa} < 1$, and $\|\widetilde{C}_0 - C_0\|_{\mathrm{HS}} \leq d$, then*

$$|D(\widetilde{C}_0) - D(C_0)| \leq \frac{d(\|C_0\|_{\mathrm{HS}} + \|\widetilde{C}_0\|_{\mathrm{HS}})}{2(1-\bar{\kappa}^2)}.$$

Proof. For $A = C_0^*C_0$ and $\widetilde{A} = \widetilde{C}_0^*\widetilde{C}_0$, the segment between them is positive and bounded by $\bar{\kappa}^2 I$. Differentiate $-\frac{1}{2}\log\det(I - A_t)$ along that segment and use the trace/operator-norm duality. The product inequality $\|X^*Y\|_1 \leq \|X\|_{\mathrm{HS}}\|Y\|_{\mathrm{HS}}$ gives $\|\widetilde{A} - A\|_1 \leq d(\|C_0\|_{\mathrm{HS}} + \|\widetilde{C}_0\|_{\mathrm{HS}})$. □

3.1 Physical Gram matrices and the missing tail

Let independent probe representers $h_i \in V_H$, $p_j \in V_p$ have Gram matrices G_H, G_p and cross matrix $X_{ij} = \langle h_i, p_j \rangle_{\mathcal{E}}$. The finite operator in orthonormal coordinates is

$$C_{\text{fin}} = G_H^{-1/2} X G_p^{-1/2}, \qquad e = \|C\|_{\text{HS}}^2 - \operatorname{tr}(G_p^{-1} X^* G_H^{-1} X). \tag{6}$$

Real transposes may replace stars throughout this report. Singular Gram matrices require removing dependencies, not an arbitrary inverse. Enclosing raw entries does not enclose (6) without lower Gram eigenvalue bounds. These are precisely the metric and omitted-space issues emphasized by the original physical-metrics chapter [3].

No list of finite probes supplies the first term in (6). For example append r mutually orthogonal singular directions of size $\kappa/2$, invisible to every listed probe. All finite data stay fixed, while the missing squared Hilbert–Schmidt norm grows as $r\kappa^2/4$. The source avoids this obstruction by its uniform harmonic singular-value tail. A usable certificate must retain that tail or another independent bound on the orthogonal complement.

4 Charged sources: the first-order term cannot be discarded

Let $f \in \mathcal{E}'$ be a real continuous load, with Riesz representer v. Let $x = \Pi_H v$, $y = \Pi_p v$, $w = (x, y)$, and on $V_H \oplus V_p$ set

$$M = \begin{pmatrix} I & C \\ C^* & I \end{pmatrix}, \qquad F(C, w) = \tfrac{1}{2}\langle w, (I - M^{-1}) w \rangle. \tag{7}$$

The spectrum of M is in $[1 - \kappa, 1 + \kappa]$.

Proposition 4.1 (Source-paired pin interaction)**.** *For the Gaussian tilted by e^f, the zero-pin interaction is $D(C) + F(C, w)$. For the normalized imaginary insertion e^{if} the algebraic density-ratio identity is $D(C) - F(C, w)$. The latter is not a positive change of measure. Both identities mean limits of finite Gaussian observations and hold for the admitted energy-continuous load.*

Proof. Synthesis $S(u, z) = u + z$ has $S^*S = M$ and $S^*v = w$, so the energy of the projection of v onto $V_H + V_p$ is $\langle w, M^{-1} w \rangle$. The range is closed because $M \geq (1 - \kappa) I$. Conditioning on a zero pin removes the corresponding projected part of the Gaussian load variance. In the logarithm of the joint zero-pin density divided by the two marginal zero-pin densities, the real tilt therefore contributes

$$\tfrac{1}{2}\{\|x\|^2 + \|y\|^2 - \|\Pi_{V_H + V_p} v\|^2\} = F(C, w).$$

The centered term is (3). Replacing f by if reverses the quadratic term by the Gaussian characteristic-function formula. Finite orthogonal probe limits converge strongly on v; the determinant limit uses (2), so no infinite pointwise density is assumed. □

Theorem 4.2 (Retained source term and signed residual)**.** *Use the projections of Theorem 3.1. Put $w_0 = (Px, Q_0 y)$, $M_0 = \left(\begin{smallmatrix} I & C_0 \\ C_0^* & I \end{smallmatrix}\right)$, $\Delta = M - M_0$, $u_0 = M_0^{-1} w$, and $r = \Delta u_0$. Define $L = \tfrac{1}{2}\langle u_0, \Delta u_0 \rangle$. Then*

$$F(C, w) - F(C_0, w_0) = L - \tfrac{1}{2}\langle r, M^{-1} r \rangle. \tag{8}$$

Consequently the error in the full real-tilted interaction lies in

$$\left[L + \frac{e}{2} - \frac{\|r\|^2}{2(1 - \kappa)}, \quad L + \frac{e}{2(1 - \kappa^2)} - \frac{\|r\|^2}{2(1 + \kappa)} \right]. \tag{9}$$

For the imaginary insertion negate the source difference in (8), rather than reuse (9).

Proof. $I - M_0^{-1}$ is supported on $\operatorname{ran} P \oplus \operatorname{ran} Q_0$, so $F(C_0, w) = F(C_0, w_0)$. The inverse identity

$$M^{-1} = M_0^{-1} - M_0^{-1}\Delta M_0^{-1} + M_0^{-1}\Delta M^{-1}\Delta M_0^{-1}$$

is verified by multiplication, or twice applying the resolvent identity. Pairing with w proves (8). The bounds $(1+\kappa)^{-1}I \le M^{-1} \le (1-\kappa)^{-1}I$, together with Theorem 3.1, give (9). □

In particular $\|\Delta\| = \|C - C_0\| \le \sqrt{e}$ and $\|u_0\| \le \|w\|/(1-\kappa)$, so the residual after retaining L is at most $e\|w\|^2/[2(1-\kappa)^3]$ in absolute value. Without computing L, the simpler resolvent estimate gives only

$$|F(C, w) - F(C_0, w_0)| \le \frac{\sqrt{e}\|w\|^2}{2(1-\kappa)^2}. \tag{10}$$

The vector u_0 includes the source's unobserved components. Thus a certificate for L, r requires source-residual bounds in addition to finite pin Gram matrices. This is a requirement, not an algorithmic oracle for unknown continuum data.

4.1 A sharp obstruction and a numerical-scale example

Take two unit energy vectors with inner product $c \in (0,1)$, pin spaces their lines, and choose v so that $x = y = 1$ in their unit coordinates. Explicitly $v = (h+p)/(1+c)$. With no probes, $C_0 = 0$, $e = c^2$, and

$$D(C) = -\tfrac{1}{2}\log(1-c^2), \qquad F(C, w) = \frac{c}{1+c}.$$

Thus the charged correction is order $\sqrt{e}$, not order e. Here $L = c$, $r = (c, c)$, and $\frac{1}{2}\langle r, M^{-1}r\rangle = c^2/(1+c)$, exactly as in the theorem. At $c = 1/4$, the source correction is $1/5$, whereas the determinant correction lies between $1/32$ and $1/30$. A determinant-only estimate would miss the larger effect even in this two-dimensional example.

For a centered computation with $\kappa = 1/2$ and a certified missing tail $e \le 10^{-4}$, the upper determinant error is $1/15000$. If the source has $\|w\| \le 1$, the coarse source bound is $1/50$; after retaining L, the residual bound is $1/2500$. These are deterministic bounds, not measured errors or a claim about the actual source's numerical constants.

5 Eliminate the source and the boundary together

The elementary Schur identity matters here because the spin insertion can be nonneutral and because Gaussian observations include interior loads. Let A be a real positive definite stiffness matrix, partitioned into interior and retained boundary coordinates:

$$A = \begin{pmatrix} A_{II} & A_{IB} \\ A_{BI} & A_{BB} \end{pmatrix}, \qquad S = A_{BB} - A_{BI}A_{II}^{-1}A_{IB}.$$

For any two real loads f, g, put $f_{\text{eff}} = f_B - A_{BI}A_{II}^{-1}f_I$, and similarly for g. Then

$$f^T A^{-1} g = f_I^T A_{II}^{-1} g_I + f_{\text{eff}}^T S^{-1} g_{\text{eff}}. \tag{11}$$

Indeed solving the interior equation gives $u_I = A_{II}^{-1}(g_I - A_{IB}u_B)$, and the boundary equation becomes $Su_B = g_{\text{eff}}$. Pairing this solution with f proves the formula. The load correction and

the local interior term are indispensable. For $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ and $f = g = (1,0)^T$, the answer is $2/3 = 1/2 + (1/2)^2/(3/2)$. Keeping only the uncorrected boundary load would give zero.

This is classical elimination, explicitly derived in the later private comparison [4]; we claim no new identity here. For a free connected graph Laplacian it applies after choosing a gauge and restricting to loads with total zero. Each free component has its own compatibility condition. A killed Laplacian instead needs no artificial neutrality: the fixed or fused boundary absorbs the missing total charge. The source uses exactly this distinction in its Villain and XY current formulas. Changing the inverse or its gauge without changing the paired load is not an admissible normalization change.

5.1 Energy-residual errors for the charged certificate

Suppose a conforming approximation $\widehat{v} \in \mathcal{E}$ to the Riesz representer of f has residual $r_f(u) = f(u) - T(\widehat{v}, u)$. By the Riesz isometry,

$$\|v - \widehat{v}\|_{\mathcal{E}} = \|r_f\|_{\mathcal{E}'}. \tag{12}$$

This includes the omitted space, not just residuals against a finite test list. If $\widehat{w} = (\Pi_H \widehat{v}, \Pi_p \widehat{v})$, then $\|w - \widehat{w}\| \le \sqrt{2}\|r_f\|_{\mathcal{E}'}$. Since $I - M^{-1} = (M - I)M^{-1}$, its norm is at most $\kappa/(1-\kappa)$. Consequently

$$|F(C, w) - F(C, \widehat{w})| \le \frac{\kappa}{2(1-\kappa)} \|w - \widehat{w}\|(\|w\| + \|\widehat{w}\|). \tag{13}$$

This supplies a separate load-error budget for Theorem 4.2. Original variational source/boundary matching in [2] and the physical-metric omitted-residual rule in [3] are relevant antecedents. Neither supplies a computable continuum dual norm without an additional residual enclosure.

6 An area-weighted cell-penalty extension

We now change the cell-mass law explicitly. Let $F \subset \mathbb{R}^2$ be a bounded connected Lipschitz domain and let $K \subset F$ have positive area. Assume that K, up to null sets, is a finite union of rectangles and admits finite rectangular partitions $\mathcal{D}_h$ whose largest side length is at most h. Partitions need not be nested. Let $\Pi_h : L^2(K) \to L^2(K)$ be the orthogonal projection onto functions constant on these cells. The forms on $H^1(F)$ are

$$T_{h,\lambda}(u, v) = a^{-1} \int_F \nabla u \cdot \nabla v + \lambda \sum_{d \in \mathcal{D}_h} |d| (\operatorname{avg}_d u)(\operatorname{avg}_d v), \tag{14}$$

$$T_{\lambda}(u, v) = a^{-1} \int_F \nabla u \cdot \nabla v + \lambda \int_K uv, \qquad a, \lambda > 0. \tag{15}$$

There is no outer Dirichlet condition. The cell penalty confines the constant. Its weights are $\lambda|d|$ on the average observations; their number increases under refinement. In contrast, the source's fixed-cell localization lemma has weights one and fixed positive cell areas. The source also uses area-weighted penalties in its earlier continuum mass construction. Our theorem concerns (14) as a continuum family, not uniformity of every fixed-cell source lemma.

Theorem 6.1 (Explicit covariance comparison)**.** *For $f \in (H^1(F))'$, write $Q_{h,\lambda}(f) = f(T_{h,\lambda}^{-1} f)$ and $Q_\lambda(f) = f(T_\lambda^{-1} f)$, where the inverse denotes the weak variational solution. With $\epsilon = a\lambda h^2/\pi^2$,*

$$\frac{Q_{h,\lambda}(f)}{1+\epsilon} \le Q_\lambda(f) \le Q_{h,\lambda}(f), \qquad 0 \le Q_{h,\lambda}(f) - Q_\lambda(f) \le \frac{\epsilon}{1+\epsilon} Q_{h,\lambda}(f). \tag{16}$$

Proof. The mean-zero Neumann Poincare inequality on a rectangle of maximum side h has constant at most h^2/π^2. One obtains it by expanding in the tensor cosine basis; every nonconstant frequency has eigenvalue at least π^2/h^2. Summing over cells gives

$$\|u - \Pi_h u\|^2_{L^2(K)} \le (h^2/\pi^2)\|\nabla u\|^2_{L^2(K)}. \tag{17}$$

Orthogonality and (17) imply $T_{h,\lambda} \le T_\lambda \le (1+\epsilon)T_{h,\lambda}$. All forms are coercive: the anchored Poincare inequality

$$\|u\|_{H^1(F)} \le C_{F,K}(\|\nabla u\|_{L^2(F)} + |\textstyle\int_K u|)$$

follows by contradiction from compactness and connectedness, since a constant with zero K-integral is zero. Moreover $|\int_K u| \le |K|^{1/2}\|\Pi_h u\|_{L^2(K)}$. Finally the variational formula $f(T^{-1}f) = \sup_u\{2f(u) - T(u,u)\}$ reverses the form inequalities and gives (16). □

The comparison is useful when λh^2 is small; it gives no small relative error otherwise. Nevertheless the hard-band limit exists with no such restriction. This distinction is useful when choosing numerical penalties.

Theorem 6.2 (Joint continuum refinement and hard-band limit)**.** *Keep F, K, a, f as above, with a fixed. Along any sequence $h \to 0$, $\lambda \to \infty$, let $u_{h,\lambda} = T^{-1}_{h,\lambda} f$. Then $u_{h,\lambda} \to u_\infty$ strongly in $H^1(F)$, where*

$$V_\infty = \{u \in H^1(F) : u = 0 \text{ a.e. on } K\}, \qquad a^{-1}\int_F \nabla u_\infty \cdot \nabla v = f(v) \quad (v \in V_\infty).$$

Also $Q_{h,\lambda}(f) \to f(u_\infty)$ and $\lambda\|\Pi_h u_{h,\lambda}\|^2_{L^2(K)} \to 0$. For every $u \in H^1(F)$ one has the quantitative leakage estimate

$$\|u\|^2_{L^2(K)} \le \left(\lambda^{-1} + \frac{ah^2}{\pi^2}\right) T_{h,\lambda}(u,u). \tag{18}$$

Proof. The last inequality is the orthogonal decomposition into cell means and their complement, followed by (17). For $\lambda \ge \lambda_0 > 0$, the anchored inequality in the preceding proof gives a coercivity constant independent of h, λ. Testing the equation with $u_{h,\lambda}$ therefore bounds both its H^1 norm and its total energy uniformly. Every subsequence has a further subsequence converging weakly in H^1 and strongly in L^2. By (18) its limit u belongs to V_∞. For $v \in V_\infty$, every cell mean of v is zero, so testing the original equation and passing to the limit gives $a^{-1}\int \nabla u \cdot \nabla v = f(v)$. The anchored inequality proves uniqueness of this solution.

Continuity of the fixed functional on weakly convergent H^1 sequences now gives

$$T_{h,\lambda}(u_{h,\lambda}, u_{h,\lambda}) = f(u_{h,\lambda}) \longrightarrow f(u_\infty) = a^{-1}\|\nabla u_\infty\|^2_2.$$

Weak lower semicontinuity, together with nonnegativity of the penalty, forces convergence of the gradient norms and vanishing of the penalty energy. Weak convergence and convergence of norms imply strong gradient convergence; compactness already gives strong L^2 convergence. Uniqueness removes subsequence extraction and proves all assertions. □

For example $\lambda = h^{-4}$ makes the relative comparison parameter diverge, but (18) is still $O(h^2)$ on bounded-energy solutions and Theorem 6.2 applies. This gives a joint continuum convergence statement, not an effective H^1 convergence rate. The proof also permits a fixed finite list of loads, by polarization. For disconnected F, every component must meet the mass region in positive area, or the unconfined constants must be separately treated.

A useful normalization check is the following. If $g \in H^1(F)$ vanishes on K and $f(v) = \int_F \nabla g \cdot \nabla v$, then $u_{h,\lambda} = ag$ for every partition and penalty. Thus the covariance is $a \int_F |\nabla g|^2$. This is the source's own weak variational normalization, not a new stiffness identity. In a smooth domain the load includes the boundary flux $\partial_n g$; replacing it by the interior function $-\Delta g$ alone changes the problem unless that flux vanishes or the tests have the corresponding boundary restriction.

6.1 What the joint limit does not exchange

The source's lattice exploration and resampling estimates use their stated order $n \to \infty$, then cell scale tending to zero, then $\lambda \to \infty$, with constants allowed to depend on the fixed penalty and geometry. Theorem 6.2 does not exchange those lattice limits. Nor does strong variational convergence provide a uniform determinant tail. For example operators equal to κ times the projection onto coordinates $j, \dots, 2j$ in ℓ^2 converge strongly to zero while their squared Hilbert–Schmidt norms grow as $(j+1)\kappa^2$. Transferring the joint continuum result to pin interactions requires uniform versions of (2), its Hilbert–Schmidt tail, and the relevant source-residual bounds.

7 Source architecture and retained obligations

All ten scientific section files were read literally in bounded displays, including the source's proofs rather than only their headings. The following map records their role, not independent validation of each proof. Filenames below refer to the cached source's `build/sections`; their full line intervals are in the read ledger.

Source section	Mechanism and boundary of this report
`introduction`	Height stiffness, roughening endpoint, and fixed low-temperature spin claims. We preserve their distinct temperature and normalization quantifiers.
`structural`	Finite Gaussian identities, reflection/monotonicity and free-boundary inputs, replica cumulants. These do not follow from a finite pin determinant certificate.
`toolkit`	Neutral, charged, wall and marked RG estimates; regulator and extraction budgets. None is replaced by our covariance inequalities.
`annular`	Cable exploration, full-cell mass penalties, weak Gaussian limits, harmonic localization, pin density ratios and conditional transfer. The local refinements target this interface; the exploration and its limit order remain inherited.
`villain`	Exact current/height duality, charge compensation at the fused boundary, low-temperature RG shooting and full-field passage. Correlations alone are not a substitute for its collision and tightness estimates.
`xy-histories`	Mixed complex covariance, periodization, live-history transports, defect tags and integrated bounds. Our positive Gaussian arguments apply only to the separately identified real coercive energy law.
`xy-flow`	Coupled tensor flow, admissible shooting, stiffness and amplitude matching. This is not proved by changing a Gram basis.
`weak-height`	Compensated weak activity, ultraviolet estimates, different entry and weak-height kernels, balanced critical flow. The two kernels retain their source scaling relation.

Source section	Mechanism and boundary of this report
`entry`	Physical observation law, isolated-core events, real comparison, complex analytic upgrade and entry at fixed temperature. A finite observation certificate does not supply its full activity norm.
`endpoint`	True-reference differentiation, replica control, shooting, critical value and endpoint slope. These remain separate global inputs.

The closest source locators are `annular.tex` 1401–1476 (weak source normalization), 1478–1888 (fixed-cell Gaussian localization and its Hilbert–Schmidt tail), 1889–2020 (full-pin mass ratios), and 2022–2355 (conditional transfer). In particular its centered conditional-density identity is not asserted unchanged under a linear tilt. Our charged correction is consistent with that warning, not a discovery that the source silently assumed a false identity. We identify no new global source contradiction here.

8 Comparison, current literature, and practical limits

The private original PDE passages [2] support keeping the boundary condition, nullspace, load and weak inverse together. The original physical-metrics chapter [3] gives the physical Gram and omitted-space viewpoint. The later comparison [4] contains an actual paired-elimination proof, but is not represented as an earlier foundational source. Their roles are methodological: they do not contain the BKT renormalisation, pin geometry, or endpoint theorems. The previous M394 kernel and fixed-hole estimates are not counted again as contributions of M397.

The fresh public primary check on October 9, 2026 was confined to abstracts and bibliographic metadata for the following comparisons. Bauerschmidt, Park and Rodriguez prove high-temperature discrete Gaussian scaling results on the torus and in infinite volume [5, 6]; those stated regimes are not the full critical roughening assertion of the cached source. Durand and Lammers [7] state the equality between XY mass and a one-sided derivative of the dual height free energy, with a quadratic error estimate near zero slope. This is not itself a proof of the source's full normalized field limit or the claimed critical stiffness. Junnila, Saksman and Webb [8] study imaginary chaos as random distributions, with moment and regularity information; identifying a particular microscopic spin limit still requires its own lattice estimates. No complete proof audit of these external papers is claimed, and no confidential title or passage was used in a public query.

The practical advantage of Theorem 3.1 is a quadratic complement error and a monotone interval for the centered interaction, including unequal pin-probe ranks. Theorem 4.2 explains exactly why a source needs more information and turns the remaining error into a signed residual after retaining its linear term. Theorems 6.1 and 6.2 separate an effective relative comparison from a rate-free continuum hard-band limit. These are local improvements using classical methods, not claims of historical priority for singular-value, Schur, or penalty arguments.

Their costs are material. Physical Gram conditioning, the spectral gap, the missing Hilbert–Schmidt tail, and source dual norms must actually be bounded. When κ approaches one the constants deteriorate. Cell refinement increases rank and changes the mass weights; fixed-cell source constants cannot simply be reused. Neither a deterministic finite certificate nor strong continuum convergence controls the critical RG trajectory, microscopic charge collisions, or distributional tightness. Useful next work would certify harmonic tails and dual residuals in a specified geometry, then establish uniformity for a specified growing family of observations. Such work would have to precede any exchange of the source's lattice limits.

9 Verification and contribution assessment

The author-owned exact script `M397-checks.py` passed 2,758 parameter cases comprising 10,962 listed assertions. It uses rational arithmetic throughout: 1,296 pin/source block cases, 19 scalar obstructions, 1,200 paired Schur cases, and 243 weighted two-mode penalty cases. Logarithms in the finite determinant fixtures are enclosed by a degree-12 rational series and a geometric remainder. The penalty fixtures test the form-comparison algebra, not continuum rectangular eigenvalues. Internal domain guards are not counted, and one whole-matrix identity counts as one assertion. Parent checks are independent and are not included in these totals.

The read ledger records 12,813 public lines across thirteen cached text files (ten scientific sections, main file, bibliography and README). It separately records 732 private lines: 235 in the original PDE report, all 210 in the physical-metrics chapter, and 287 in the later paired-source comparison. Unread complements of the private files are explicit. Pre-read hashes were persisted before those scientific displays and matched the post-read endpoint hashes. These facts do not constitute an exhaustive private-corpus audit or a global proof certification. The source subtree's captured-commit provenance is also distinct from mathematical verification.

The contribution rating is C3, independently of the existing L3 applicability classification. It rests on the explicit centered and charged pin certificates, their obstruction/example, and the carefully weighted joint continuum extension. Classical identities, successful finite tests, public-source provenance and typesetting quality are not themselves credited as new mathematical contributions. The resulting report gives quantitative, conditional Gaussian tools and a proved continuum extension, not a new proof of BKT universality.

References

[1] *BKT universality for height and planar spin fields*, September 24, 2026, cached public manuscript. Pinned public source. All ten scientific section files read; external references are separate.

[2] M. Subbarao, original private PDE manuscript, `TR8-lin-non-lin.tex`, local confidential corpus. Fresh passages: lines 53–190, 492–558, 1008–1037.

[3] M. Subbarao, original physical-metrics chapter, Ram Calculus books, volume 21, `ch01d-physical-metrics.tex`, lines 1–210. Local confidential corpus.

[4] M. Subbarao, later private comparison `TR-meta5-math.tex`, lines 1–95, 154–202, 454–596. Local confidential corpus.

[5] R. Bauerschmidt, J. Park and P.-F. Rodriguez, *The Discrete Gaussian model, I. Renormalisation group flow at high temperature*, Ann. Probab. 52 (2024), 1253–1359. Primary abstract and metadata.

[6] R. Bauerschmidt, J. Park and P.-F. Rodriguez, *The Discrete Gaussian model, II. Infinite-volume scaling limit at high temperature*, Ann. Probab. 52 (2024), 1360–1398. Primary abstract and metadata.

[7] T. Durand and P. Lammers, *The BKT transition and surface tension differentiability*, May 27, 2026. Primary abstract and metadata.

[8] J. Junnila, E. Saksman and C. Webb, *Imaginary multiplicative chaos: Moments, regularity and connections to the Ising model*, arXiv version 2, December 20, 2018. Primary abstract and metadata.